

Scaling Whole-Body Loco-Manipulation through Compositional Data Synthesis

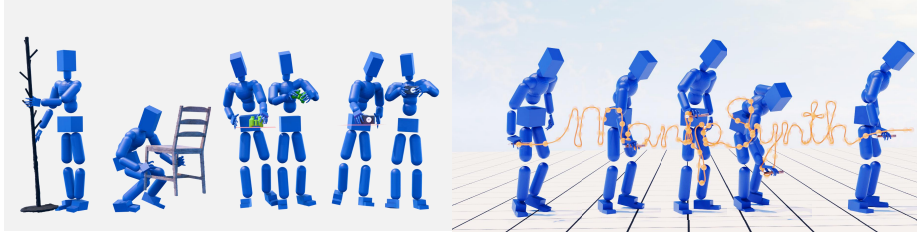


Fig. 1: Our method synthesizes loco-manipulation behaviors between humanoids and diverse objects, including **(Left)** interactions with large and small objects, single-handed grasping, bimanual handovers, as well as **(Right)** long-horizon object transportation and orientation.

Abstract. Acquiring diverse whole-body dexterous manipulation data for humanoids has long been a fundamental challenge in character animation and robotics. Existing methods either rely on costly motion capture systems or synthesize motions with poor physical plausibility, hindering the acquisition of high-quality, diverse datasets at scale. In this work, we present **ManipSynth**, a unified and morphology-agnostic framework for scalable whole-body loco-manipulation data synthesis. ManipSynth employs a compositional synthesis strategy: whole-body motion is generated via learning-based sparse whole-body tracking, while hand-object motion and contact transitions are synthesized through contact-aware, rule-based optimization followed by physical refinement. Using the data generated by ManipSynth, we further train **ManipTracker**, a general-purpose Humanoid-Object Interaction (HOI) tracking controller capable of steering physically simulated humanoids to imitate the reference HOI data. Experiments demonstrate that ManipSynth achieves approximately $5\times$ higher interaction coverage and $6\times$ lower penetration than OMOMO across a range of representative objects. Meanwhile, ManipTracker attains around 90% tracking success rate on our synthetic dataset and maintains strong generalization performance. Code for our data synthesis pipeline and controller training will be released upon paper acceptance.

Keywords: Whole-body loco-manipulation · Human-object interaction
· Synthetic data generation

1 Introduction

Humans perform loco-manipulation with the *whole body*, not the hands alone. Actions such as picking up, placing handing over, and carrying objects require coordinated whole-body balance, body-object contacts, and object-aware motion.

Modeling these whole-body behaviors is fundamental to physics-based character animation and robotics, yet most existing manipulation pipelines are primarily built for robotic-arm grippers or dexterous hands [3, 12–14, 23, 33, 40, 41, 44, 45]. This motivates us modeling manipulation from a whole-body perspective.

Recent approaches [11, 15, 28, 32, 34–36, 38, 39, 42] have made significant progress in learning physically plausible human object interactions (HOI). These methods mainly fall into two paradigms: (1) whole-body prior-based control with task-specific rewards and (2) human demonstration-based learning with imitation rewards. The first paradigm can learn robust whole-body grasping skills [1, 4, 17], but each new interaction task typically requires redesigning new rewards. The second paradigm relies on human demonstrations [28, 32, 36, 38, 42], either captured via motion capture systems or reconstructed from videos. However, motion capture typically requires specialized hardware and extensive manual effort [9, 16, 26] to obtain high-quality data, which limits scalability across diverse object categories, interaction modes, and environments. Although video-reconstructed demonstrations [2, 8, 35, 36] could scale more easily, they are often prone to noises, particularly in hand-object contact and object pose estimation.

Different from existing approaches, synthetic data generation provides an efficient and much more scalable alternative [5–7, 30, 33, 46], while enabling stronger quality through physically grounded generation constraints. Although existing synthetic data pipelines are numerous, they all target single-arm or dual-arm grippers [5–7, 30] or dexterous hands [33, 46], and solely consider table-top manipulation scenarios. Therefore, how to synthesize physically plausible whole-body loco-manipulation data remains largely unexplored.

In fact, while existing data synthesis methods have achieved significant progress in dexterous hand and robotic arm manipulation, extending such synthesis paradigms to whole-body dexterous interaction poses three challenges: First, the human whole body contains over 20 joints, making direct trajectory planning for each joint inefficient and impractical in such a high DOF space. A feasible strategy is to plan trajectories for only 2–3 key joints (such as root and wrists) and generate coherent full-body motion from these sparse signals. Second, as shown in Table 2, tracking wrist-only trajectories yields substantially larger errors, therefore, sparse tracking alone is insufficient for precise hand-object coordination. Finally, directly concatenating frame-wise whole-body poses and hand poses can produce plausible static snapshots, but when rolled out as continuous motion, the resulting trajectories often exhibit physical artifacts, such as foot sliding and hand-object interpenetration, which require further correction for physical plausibility.

To address the above challenges, we propose ManipSynth, a morphology-agnostic, scalable, and unified framework for whole-body loco-manipulation data synthesis. The whole body motion is obtained through a sparse goal-conditioned controller (eg, PULSE [18] and MaskedMimic [27]), while stable grasp poses are generated via force-closure optimization [31] to ensure physically consistent contact configurations. Continuous interaction sequences are synthesized by composing transitions between carefully constructed key interaction states. By interpo-

lating and stitching these states, we generate structured pseudo-demonstrations that approximate realistic loco-manipulation trajectories at scale. To address physical implausibilities in pseudo-demonstrations, we further apply HOI tracking methods [28, 38, 42], which can tolerate moderate contact and kinematic inconsistencies, allowing imperfect synthetic data to provide effective supervision. Leveraging ManipSynth, we focus on four representative interaction categories: single-hand manipulation of small objects, bi-manual handover of small objects, single-hand manipulation of large objects, and bi-manual manipulation of large objects, which involve varying contact configurations, load distributions, and coordination strategies. ManipSynth achieves **5× higher** average interaction coverage than OMOMO and **6× lower** penetration, indicating better data efficiency and contact quality. Leveraging the synthetic HOI trajectories produced by ManipSynth, we train a general-purpose ManipTracker using skill-agnostic human-object interaction imitation rewards. Specifically, instead of optimizing for task-specific objectives, our tracker is trained to imitate human-object states, enabling it to serve as a reusable low-level interaction controller across different manipulation scenarios. ManipTracker achieves close to 90% tracking success rate on training data and maintains strong performance on the test split, demonstrating effective generalization from purely synthetic training data.

In summary, our contributions are threefold.

1. We propose ManipSynth, which, to the best of our knowledge, is the first unified, scalable, and morphology-agnostic framework for whole-body loco-manipulation data synthesis under a unified formulation.
2. We establish a new, large-scale, and continually expandable 4D HOI dataset via ManipSynth. This dataset enables learning a generalizable low-level HOI tracking primitive for high-dynamic whole-body manipulation.
3. Leveraging ManipSynth synthetic demonstrations, we develop ManipTracker, a general-purpose HOI tracking controller trained with task-agnostic imitation rewards and Physical State Initialization for improved motion naturalness.

2 Related Work

2.1 Demonstration Data for Loco-Manipulations

Demonstration data for loco-manipulations has evolved from controlled motion-capture datasets to large-scale and in-the-wild resources. Mocap-based HOI datasets, such as GRAB [26], OMOMO [9], and HUMOTO [16], provide high-quality body-object trajectories and contact details, but they are costly to collect and difficult to deploy in portable, large-scale settings. Beyond mocap, reconstruction-based resources such as BEHAVE [2], InterCap [8], Open3DHOI [35] and Open4DHOI [36] improve scene diversity and real-world coverage, but still suffer from multi-view RGB or RGB-D estimation errors in hand-object contacts and object poses, as well as severe occlusion in human-object interaction regions.

Table 1: Comparison with representative physics-based HOI methods. We compare scalability of data generation, reliance on demonstrations, morphology generality, and data sources.

Method	Year	Scalable Data	Demonstration-Based	Morphology-Agnostic	Data Sources
PhysHOI [32]	2023	✗	✓	✗	Mocap
PMP [1]	2023	✗	✗	✗	No
Braun et al. [4]	2024	✗	✗	✓	No
Li et al. [10]	2025	✗	✓	✗	Mocap
Omnigrasp [17]	2024	✗	✗	✓	No
InterMimic [38]	2025	✗	✓	✗	Mocap
SkillMimic-V2 [42]	2025	✗	✓	✗	Mocap
MaskedManipulator [28]	2025	✗	✓	✗	Mocap
Wen et al. [36]	2025	✓	✓	✗	Video
ManipTracker (Ours)	2026	✓	✓	✓	Synthetic

Another data-collection paradigm uses high-density dome capture systems, such as NeuralDome pipelines [43], to improve reconstruction fidelity; however, these setups require dense camera infrastructure and incur substantial hardware and operational cost. Taken together, these three acquisition paradigms face a common triad of bottlenecks: high data cost, reconstruction noise with occlusion, and heavy hardware dependence. Theoretically, purely synthetic data is substantially cheaper to produce and easier to scale up, yet principled rule-driven synthesis of physically grounded HOI trajectories remains largely underexplored.

2.2 Reinforcement Learning for Loco-Manipulations

As shown in Table 1, recent physics-based RL methods for HOI can be broadly grouped into goal-conditioned whole-body grasping and data-driven interaction imitation. Methods such as PMP [1], physically plausible full-body HOI synthesis [4], and Omnigrasp [17] typically decouple strong whole-body motion priors from interaction optimization, and then solve target behaviors with explicitly designed task rewards. Although this design can generalize well in grasping-centric settings, it does not readily scale to broader and more compositional loco-manipulation skill sets, such as bimanual handover. In contrast, data-driven interaction imitation methods, including PhysHOI [32], SkillMimic [34], SkillMimic-V2 [42], and InterMimic [38], learn controllers by imitating demonstration trajectories and show strong performance on dynamic interactions, providing a low-level control foundation for higher-level goal planning [28, 39]. However, these methods still largely depend on existing mocap-derived demonstrations, whose scale and diversity are limited by collection cost, making it difficult to scale toward broader object categories and skill families.

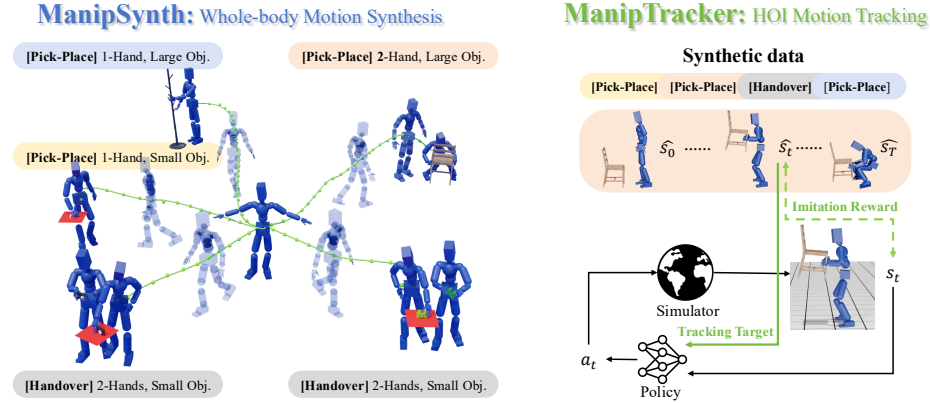


Fig. 2: Overview of ManipSynth and ManipTracker for scalable whole-body loco-manipulation. Left: ManipSynth generates diverse whole-body loco-manipulation motions across different interaction modes. Right: ManipTracker learns an HOI tracking policy in simulation from the purely synthetic demonstrations.

3 Method

3.1 Overview

We formulate **ManipSynth** as a unified, shape-agnostic framework for synthesizing whole-body loco-manipulation trajectories for human-object interaction learning. Given sparse task specifications, **ManipSynth** generates whole-body motion trajectories that achieve target interactions while enforcing contact feasibility and physical consistency under physics constraints. The framework follows a compositional synthesis paradigm consisting of three components: **Sparse Whole-Body Tracking**, **Contact-Aware Hand Motion Synthesis**, and **Physical Refinement**. At its core, whole-body motion is generated by a learning-based sparse tracking policy, while hand motion and hand-object contact transitions are synthesized through rule-based construction, combining the scalability of learned whole-body generation with the precision and controllability of rule-based hand interaction synthesis. Based on the synthesized trajectories from ManipSynth, we train **ManipTracker**, a general-purpose controller with a unified imitation reward across different skills. ManipTracker can track coupled human-object motion targets for diverse loco-manipulation behaviors, including single-hand manipulation, bimanual transport, and handover. As a reusable low-level interaction primitive, it reduces task-specific reward redesign and provides a strong foundation for higher-level planning and generative controllers [28, 39]. Figure 2 provides an overview of ManipSynth and ManipTracker, and Figure 3 presents the pipeline of the ManipSynth.

3.2 Sparse Whole-Body Tracking

Directly planning trajectories for all human joints is impractical for whole-body loco-manipulation due to the high-dimensional body DoFs. A feasible strategy is to first plan trajectories for only 2–3 key joints (e.g., root and wrists, with target positions and rotations), and then synthesize the whole-body joint positions and rotations from these sparse signals. Instead of adopting a two-stage pipeline that first reconstructs a full kinematic pose from sparse joint trajectories [24] and then performs dense tracking [19], we directly formulate the problem as sparse whole-body tracking. The intermediate reconstruction step in two-stage approaches inevitably introduces approximation errors that accumulate during dense tracking, while also imposing unnecessary supervision on joints that are irrelevant to the task. By conditioning the partially conditioned controller π^{pc} directly on task-relevant sparse signals [18, 27, 37], a unified control interface is enabled for whole body loco-manipulation data synthesis. We keep the first-stage whole-body tracking [22] training unchanged, and modify only the second-stage distillation process. In distillation, we apply joint-level randomization over three conditioning joints, namely the root, left wrist, and right wrist. At the temporal level, each condition includes the reference state at the current frame and a short look-ahead window of five future frames. This design yields a unified sparse control interface across four interaction scenarios: single-hand manipulation uses one active wrist trajectory, bimanual handover switches the active wrist trajectory online, bimanual large-object manipulation uses both wrist trajectories, and approach or retreat phases use only the root trajectory. As a result, one sparse controller can cover diverse loco-manipulation phases without task-specific controller redesign.

3.3 Contact-Aware Hand Motion Synthesis

Whole-body interaction data synthesis can be viewed as a composition of *whole-body trajectory generation* and *precise hand trajectory synthesis*. The sparse tracking controller in Sec. 3.2 mainly handles the former, while this module focuses on the latter by constructing reliable hand-object contact states. We first synthesize object-centric grasp poses using force-closure optimization [31], which provides force-closure-compatible hand-object configurations as contact anchors. After grasp synthesis, we apply quality filtering before trajectory generation: for small objects, candidate grasps are filtered by both penetration and gravity-resistance criteria, for large objects, we retain grasps using penetration-based filtering only. Using these contact anchors, we synthesize data of three common daily interaction skills: single-hand pick-and-place, bimanual pick-and-place, and bimanual handover. For all three skills, we first plan a root trajectory that brings the human close to the object; after this approach stage, the synthesis proceeds with the following three skill-specific cases.

Single-hand pick-and-place. Single-hand pick-and-place synthesis consists of a grasping transition from no contact to in contact, a transport phase that preserves the relative hand-object pose after grasp establishment, and a placement

transition from in contact back to no contact. We first construct a single-wrist trajectory for the full process (grasp + free move + place), where the wrist trajectory can be provided by a high-level planner or synthesized via randomization [25]. Conditioned on this wrist trajectory, we use π^{pc} trained in Sec. 3.2 to generate whole-body joint trajectories for the entire sequence. Finger poses are generated by linearly interpolating joint angles from the zero hand pose to the grasp pose at contact establishment. Concatenating the whole-body trajectory with the synthesized finger poses yields the final whole-body motion trajectory.

Bimanual pick-and-place. Compared with single-hand pick-and-place, bimanual pick-and-place requires planning two wrist trajectories for the full sequence, including synchronized grasping, cooperative free move, and coordinated placement. The two wrist trajectories can be generated by a high-level planner or synthesized with randomized motion priors, but they remain coupled through a shared object-centric contact frame throughout the free-move phase. In the free-move stage, we explicitly constrain the relative transform between the two wrists to remain unchanged, so that both hands preserve the same cooperative support pattern established at grasp time. Despite this coupling constraint, directly tracking these dual-wrist references with π^{pc} introduces small but accumulative tracking errors, which can break wrist-to-object alignment and degrade two-hand coordination over time. To address this issue, we perform arm-level IK alignment for both arms in the free-move phase: at each frame, IK corrects wrist positions and orientations toward the planned dual-wrist targets while preserving kinematic feasibility. This post-alignment step enforces consistent alignment with pre-synthesized grasp poses and guarantees frame-wise synchronization between the two wrists throughout transport and placement.

Bimanual handover. Similar to bimanual pick-and-place, we first pre-synthesize left-hand and right-hand grasp poses for the object. Here, we use the case of grasping with the left hand and handing over to the right hand as an example. After the human reaches a pre-grasp location in front of the object, we set the left-wrist target pose at the grasp frame and solve IK to obtain the corresponding target arm joint angles, then generate the left-arm joint trajectory by interpolation. After the left hand establishes contact, we first plan an object trajectory to a canonical handover position about 30 cm in front of the chest. Given the object pose at each frame along this trajectory, we compute the corresponding frame-wise left-wrist targets and solve IK frame by frame to obtain left-arm joint angles during object transport. This process directly produces the left-arm joint trajectory for the free-move phase. With the left arm fixed, we set the right-wrist grasp target and solve IK for the corresponding target right-arm joint angles, then interpolate to form the receiving motion. Finally, the left hand releases the object by interpolating finger pose from grasp to open, while the right hand maintains grasp and moves backward by 20 cm to complete the handover separation.

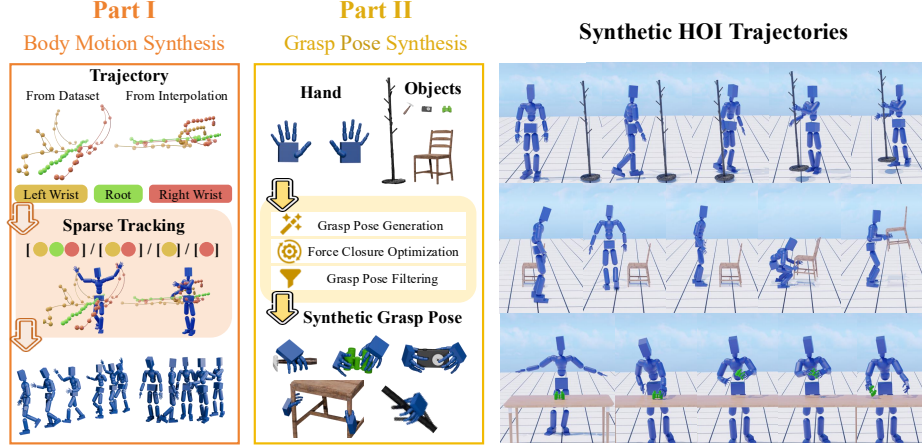


Fig. 3: Pipeline of ManipSynth for scalable whole-body loco-manipulation data synthesis. Part I constructs whole-body motion sequences by defining sparse control trajectories (e.g., root and wrist paths) from motion data or simple interpolation, then reconstructing whole-body motions via sparse body tracking. Part II synthesizes object-aware grasp poses using contact optimization [31], followed by grasp filtering. Combining the synthesized motions and grasps produces large-scale synthetic HOI trajectories spanning diverse locomotion and interaction scenarios, enabling downstream HOI tracking.

3.4 Physical Refinement

Although Sec. 3.3 provides scalable pseudo-demonstration synthesis, the generated motions still contain physically implausible artifacts. First, interpolating between initial finger states and grasp frames may introduce temporary hand-object interpenetration. Second, for large-object pick-and-place (e.g., chairs), the planned object and wrist trajectories can be overly stiff and less natural. Third, when π^{pc} generates whole-body motion from sparse wrist conditions, out-of-distribution wrist trajectories may induce foot skating. To improve robustness against such imperfect demonstrations, we follow the spirit of prior robust imitation pipelines [42]. Concretely, we combine reference-motion perturbation at initialization (data augmentation) with improved training initialization strategies, including replacing naive random-state initialization with physically valid state initialization, so the policy is trained to recover from noisy and inconsistent demonstrations while preserving task-relevant interaction outcomes. To further mitigate foot skating, we additionally adopt the built-in `feet_air_time` reward term from Isaac Gym [21]. This reward explicitly accounts for foot contact timing by modeling air-time and touchdown events, encouraging cleaner swing phases and reducing dragging or sliding artifacts during tracking.

3.5 ManipTracker

We control a humanoid with 51 actuated joints; the action space is $\mathbb{R}^{51 \times 3}$. We model ManipTracker as a goal-conditioned MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{T}, \gamma)$, where the policy $\pi_\theta(\mathbf{a}_t \mid \mathbf{o}_t, \mathbf{g}_t)$ tracks coupled human-object trajectories under contact dynamics. Given synthetic demonstrations $\tau = \{(\hat{\mathbf{x}}_t^h, \hat{\mathbf{x}}_t^o)\}_{t=1}^T$, our policy predicts residual joint targets over the humanoid DoFs:

$$\mathbf{a}_t \sim \pi_\theta(\cdot \mid \mathbf{o}_t, \mathbf{g}_t) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{o}_t, \mathbf{g}_t), \text{diag}(\boldsymbol{\sigma}_\theta^2)), \quad (1)$$

$$\mathbf{q}_t^{\text{cmd}} = \hat{\mathbf{q}}_t^{\text{ref}} + \Delta \mathbf{q}_t, \quad \Delta \mathbf{q}_t = \mathbf{a}_t, \quad (2)$$

where $\hat{\mathbf{q}}_t^{\text{ref}}$ is the synthesized reference pose and $\mathbf{q}_t^{\text{cmd}}$ is executed by the low-level PD controller in simulation.

The policy observation is decomposed as

$$\mathbf{o}_t = [\mathbf{o}_t^h, \mathbf{o}_t^o], \quad (3)$$

where $\mathbf{o}_t^h$ is the humanoid state feature, and $\mathbf{o}_t^o$ is the object-centric feature. In the root-heading local frame, we use

$$\mathbf{o}_t^h = [h_t^{\text{root}}, \mathbf{y}_t^{b, \text{loc}}, \mathbf{c}_t^h, \boldsymbol{\delta}_t^h], \quad (4)$$

where $\mathbf{y}_t^{b, \text{loc}} = [\mathbf{p}_t^{b, \text{loc}}, \mathbf{r}_t^{b, \text{loc}}, \mathbf{v}_t^{b, \text{loc}}, \boldsymbol{\omega}_t^{b, \text{loc}}]$ collects local body position, rotation, linear velocity, and angular velocity, and $\boldsymbol{\delta}_t^h = [\Delta \mathbf{p}_t^b, \Delta \mathbf{r}_t^b, \Delta \mathbf{v}_t^b, \Delta \boldsymbol{\omega}_t^b]$ collects human-state residuals to references. Here $\mathbf{c}_t^h$ denotes current whole-body joint contact forces. Thus $\mathbf{o}_t^h$ includes root height, local body kinematics, current contact forces, and compact residual terms. The object branch is

$$\mathbf{o}_t^o = [\mathbf{y}_t^{o, \text{loc}}, \mathbf{k}_t^{o, \text{loc}}, \boldsymbol{\delta}_t^o], \quad (5)$$

where $\mathbf{y}_t^{o, \text{loc}} = [\mathbf{p}_t^{o, \text{loc}}, \mathbf{r}_t^{o, \text{loc}}, \mathbf{v}_t^{o, \text{loc}}, \boldsymbol{\omega}_t^{o, \text{loc}}]$ collects local object position, rotation, linear velocity, and angular velocity, and $\mathbf{k}_t^{o, \text{loc}}$ denotes a local object geometry descriptor represented by transformed object keypoints in the root-heading local frame, and $\boldsymbol{\delta}_t^o = [\Delta \mathbf{p}_t^o, \Delta \mathbf{r}_t^o]$. This branch includes local object pose and velocity features, transformed object keypoints, and compact object residual terms. The target observation from reference motion is defined as

$$\mathbf{g}_t = [\hat{\mathbf{s}}_t, \hat{\mathbf{s}}_{t+1}, \dots, \hat{\mathbf{s}}_{t+H}], \quad \hat{\mathbf{s}}_\tau = [\hat{\mathbf{x}}_\tau^h, \hat{\mathbf{x}}_\tau^o, \hat{\mathbf{c}}_\tau^h, \hat{\mathbf{c}}_\tau^o], \quad (6)$$

where $\hat{\mathbf{x}}_\tau^h$ denotes the whole-body human reference state (global or root state, body or joint poses, and motion), $\hat{\mathbf{x}}_\tau^o$ denotes object reference state, and $\hat{\mathbf{c}}_\tau^h, \hat{\mathbf{c}}_\tau^o$ denote human or object contact indicators. We use the current reference together with a short look-ahead window H to provide target-motion context for whole-body HOI tracking. For distribution compaction and numerical stability, we represent all kinematic features in the root-heading local frame, encode rotations

with tangent-normalized quaternion features, and apply norm clipping to object-relative positions and transformed object keypoints (bounded to a fixed radius) before concatenating them into $\mathbf{o}_t$.

Unified Human-Object Imitation Reward. We adopt a unified HOI imitation reward that jointly tracks whole-body human state and object state while respecting interaction-specific constraints. Following recent HOI tracking formulations, we combine human pose tracking, object pose tracking, interaction consistency (human-object relative geometry), and contact consistency into a single scalar objective with multiplicative aggregation [33, 38]:

$$r_t = r_t^h \cdot r_t^o \cdot r_t^i \cdot r_t^c \cdot r_t^f, \quad (7)$$

where r_t^h is the human tracking reward, r_t^o the object tracking reward, r_t^i the interaction reward, r_t^c the contact reward, and r_t^f the feet-in-the-air time reward. Concretely, we target human and object state sequences from demonstrations and compute discrepancies between the agent state and the reference, including positional errors, orientation deviations, relative human-object geometry, and contact consistency. We exclude velocity terms in the imitation targets for robustness to imperfect synthetic references, and the feet air-time term discourages dragging to improve locomotion realism. Compared with task-specific reward engineering, this unified formulation generalizes across grasping, transport, and in-hand manipulation. Additional implementation details for sub-reward computation are provided in the appendix.

4 Experiments

4.1 Experimental Setup

Training Settings. We train two types of policies. π^{pc} is trained in the ProtoMotions framework [29] using joint-trajectory targets from two sources: (i) randomized trajectory generators (Trace and Pace [25]), which provide single-joint motion trajectories, and (ii) AMASS training data, where we extract trajectories for 2–3 joints from full-body motions. π^{pc} is trained for two days on 8 A100 GPUs. *ManipTracker* is trained on our synthetic loco-manipulation dataset for HOI tracking, using a single RTX 4090 GPU for 5000 epochs. Additional training settings can be found in the appendix.

Simulator Settings. We implement ManipSynth and ManipTracker in Isaac Gym [21] with consistent simulation settings across all experiments. In Isaac Gym, the physics timestep is set to 1/60 s with 2 substeps and the control frequency is 30 Hz via control Frequency is set to 2 to match the 30 FPS motion data. We use PD joint control, set the plane friction to (0.9, 0.9) with restitution 0.7, and cap the maximum depenetration velocity at 100 to improve contact stability. Additional implementation details and simulation hyperparameters are summarized in the appendix.

Dataset. We use ManipSynth to generate four long-horizon, everyday interaction categories: single-hand pick-and-place of small objects, single-hand

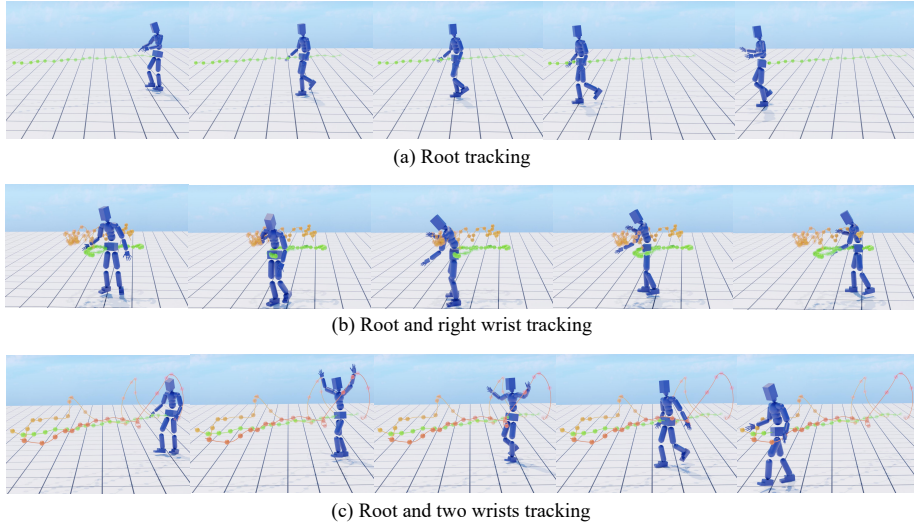


Fig. 4: Qualitative results of sparse whole-body tracking with different conditioning joints.

pick-and-place of large objects, bimanual pick-and-place of large objects, and bimanual handover of small objects. For each object, we synthesize about 100 diverse trajectories with varied grasp poses and distinct free-move path patterns.

Metrics. To evaluate the performance of ManipTracker, we report Success Rate, Human Tracking Error, Object Tracking Error, and Tracking Length. Success Rate measures the fraction of reference sequences for which the policy succeeds at least once, averaged across all references. Human and Object Tracking Errors are reported as position and rotation errors between simulated and reference trajectories (averaged over the best-performing trial). Tracking Length is the duration (in seconds) that the imitation remains stable before triggering the interaction termination conditions in Sec. 3.2.

4.2 Sparse Whole-Body Tracking Results

We adopt sparse whole-body tracking by conditioning the controller on task-relevant joint trajectories, while keeping the first-stage full-body tracking training unchanged and modifying only the second-stage distillation. In distillation, we randomize the conditioning joints over the root, left wrist, and right wrist, and each condition includes the current reference state plus a short look-ahead window. Both two stages are trained on AMASS dataset [20]. This design is motivated by two considerations: whole-body motion is difficult to synthesize reliably with rule-based methods, and synthetic single-hand/bimanual data benefit from a unified model that can track either one or both wrists, while root tracking provides a more stable gait and improves robustness across phases. Quantitatively,

Table 2: Sparse whole body tracking results (*data cleaning) on AMASS train and test (10595 and 137 sequences).

Joints	AMASS-Train*					AMASS-Test*				
	Succ $\uparrow$	$E_{g-mpjpe}$ $\downarrow$	E_{mpjpe} $\downarrow$	E_{acc} $\downarrow$	E_{vel} $\downarrow$	Succ $\uparrow$	$E_{g-mpjpe}$ $\downarrow$	E_{mpjpe} $\downarrow$	E_{acc} $\downarrow$	E_{vel} $\downarrow$
Root	95.51%	59.88	0	6.95	6.14	63.50%	219.60	0	11.71	14.10
Left wrist	93.29%	66.36	92.43	13.27	18.03	53.28%	201.16	225.49	27.05	27.70
Right wrist	92.43%	66.67	101.50	12.66	16.64	51.82%	214.26	252.19	25.34	25.44
Left wrist + Root	97.74%	49.97	15.78	8.60	10.92	78.83%	179.07	42.96	17.52	17.69
Right wrist + Root	97.67%	49.77	16.24	8.22	10.04	78.83%	142.95	34.75	17.06	16.60
Two wrists	97.50%	51.52	48.18	11.44	15.01	67.15%	204.07	123.76	24.27	24.94
Two wrists + Root	98.09%	47.60	20.70	9.06	11.31	81.75%	163.60	49.61	18.28	18.47

Table 3: Comparison of evaluation metrics between OMOMO and ManipSynth on three objects.

Evaluation metrics	OMOMO				ManipSynth				
	Interaction Coverage	Foot Sliding [mm]	Penetration [mm]	Jerk m/s^3	Interaction Coverage	Foot Sliding [mm]	Penetration [mm]	Jerk m/s^3	Time [s]
Woodchair	4.15%	1.83	4.601	60.33/30.03	20.43%	3.03	2.251	301.82/17.20	5.4
Clothesstand	3.94%	2.60	14.502	56.91/29.30	16.01%	1.72	0.058	270.40/8.88	5.2
Targetable	3.78%	3.20	6.101	59.71/27.47	19.05%	1.93	1.864	264.93/27.47	5.5
Average	3.96%	2.54	8.401	58.98/28.93	18.50%	2.23	1.391	279.05/17.85	5.37

when the tracked joints include the root and at least two joints, success exceeds 95% on AMASS-Train and approaches or surpasses 80% on AMASS-Test. In contrast, tracking a single joint or only the two wrists degrades performance, since many training motions cannot be well represented by such sparse signals. Qualitatively, we visualize three representative sequences to highlight stable gait, coordinated upper-body motion, and smooth phase transitions under sparse control, showing that the sparse controller preserves whole-body coherence while remaining flexible across interaction phases.

4.3 Synthetic Data Results

We synthesize four common interaction modes in daily life: (i) single-hand table-top pickup of a small object followed by placement or handover to the other hand, (ii) single-hand relocation of a large object, and (iii) bimanual relocation of a large object. Qualitative examples of the synthesized sequences are shown in Fig. 5, and quantitative metrics are reported in Table 3.

Across individual objects, our synthesized datasets achieve $5\times$ higher interaction coverage than OMOMO and yield $6\times$ lower penetration. The lower penetration is primarily due to using force-closure optimization [31] to generate grasp poses and filtering for contact validity. In contrast, our foot sliding and jerk metrics are weaker than OMOMO, which we attribute to residual jitter and foot-skate artifacts when π^{pc} reconstructs whole-body motion from sparse joint trajectories. The computation details for these evaluation metrics are provided in the appendix.

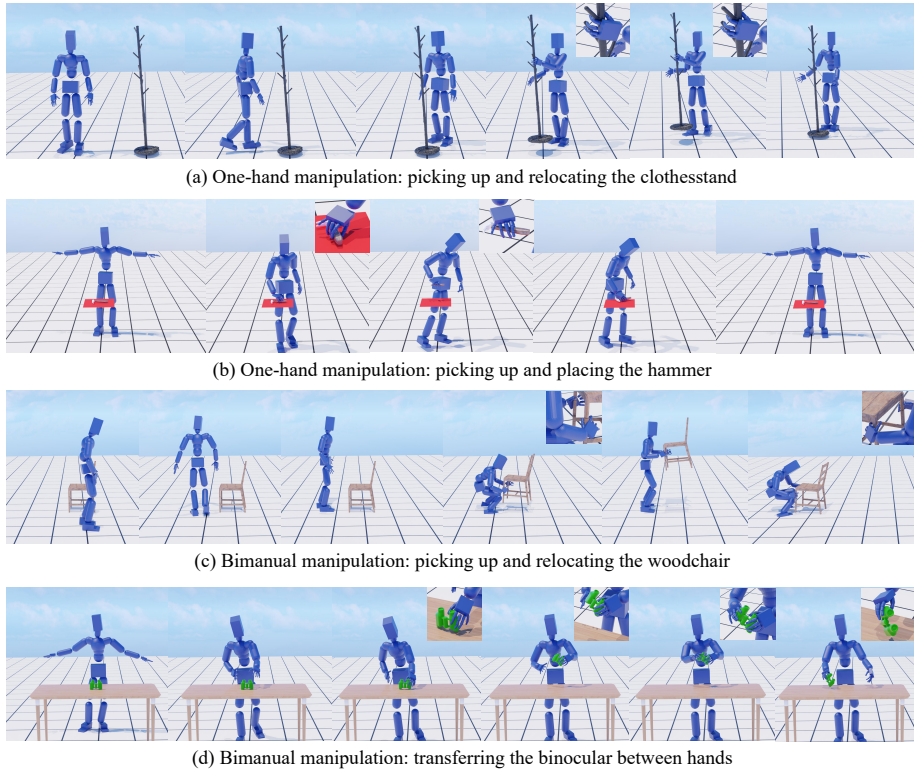


Fig. 5: Qualitative samples of synthetic loco-manipulation from ManipSynth.

4.4 HOI Tracking Performance and Ablation Study

We select three representative objects (Woodchair, Clothesstand, and Table). For each object, ManipSynth generates 120 trajectories, where 100 are used for training and the remaining 20 for testing. After data filtering, the effective trajectory counts are 77/16 (train/test) for Woodchair, 87/18 for Clothesstand, and 83/16 for Table.

As shown in Table 1, ManipTracker achieves sequence success rates close to 90% in most reported settings on both training and test splits. These results validate the effectiveness of our synthesized data for learning object-carrying skills, and also suggest promising generalization from purely synthetic training data, which we attribute to the high interaction-space coverage of our synthesized HOI trajectories.

For ManipTracker, we adopt a unified HOI imitation reward following prior imitation-based HOI trackers [28, 38, 42]. To further mitigate foot-skating artifacts, we additionally apply a feet-in-the-air-time task reward on the human body states. The corresponding reward ablations are reported in Table 5. We train on synthesized manipulation data from two objects with different opera-

Table 4: HOI Tracking Performance on synthetic data of three different objects.

Evaluation metrics	Train				Test			
	Sequence Success	Object Error [mm]	Tracking MPJPE [mm]	Tracking Length [s]	Sequence Success	Object Error [mm]	Tracking MPJPE [mm]	Tracking Length [s]
<i>Woodchair</i>	98.70%	30.0	54.1	13.2	62.50%	38.2	121.0	10.6
<i>Clothesstand</i>	95.40%	55.3	57.7	10.7	91.67%	83.9	68.1	10.5
<i>Table</i>	96.39%	65.7	54.5	13.1	80.00%	75.2	106.1	11.8
<i>Average</i>	96.83%	50.3	55.4	12.3	78.06%	65.8	98.4	11.0

Table 5: Ablation study on different reward components for HOI tracking on two different objects.

Reward	<i>Woodchair</i>				<i>Clothesstand</i>			
	Sequence Success	Object Error [mm]	Tracking MPJPE [mm]	Tracking Length [s]	Sequence Success	Object Error [mm]	Tracking MPJPE [mm]	Tracking Length [s]
w/o $r^{contact}$	64.94%	52.5	84.2	12.1	48.28%	92.5	86.6	9.4
w/o $r^{interact}$	32.47%	54.7	113.4	11.2	78.16%	95.1	116.5	10.2
w/o r^{object}	76.62%	41.2	87.7	12.1	71.26%	84.9	103.9	9.9
w/o r^{feet}	0.0%	35.3	103.5	8.0	50.57%	91.8	104.4	9.6
ManipTracker	87.01%	39.0	82.5	12.7	89.70%	70.9	76.1	10.5

tion patterns (Woodchair and Clothesstand). When all reward terms are enabled, the performance is the best. Removing r^{feet} causes a large drop on Woodchair (down to 0.0% success), where the policy frequently becomes unstable and slips during chair carrying. Removing contact supervision ($r^{contact}$) nearly halves performance on Clothesstand, and under noisy synthetic references, this often leads to failed pole grasping. More qualitative ablation results are in the Appendix.

5 Conclusion

We presented **ManipSynth**, a unified, scalable, and morphology-agnostic framework for synthesizing whole-body loco-manipulation data. By composing learning-based sparse whole-body tracking with contact-aware hand-object motion synthesis and physical refinement, ManipSynth generates diverse HOI trajectories with improved contact fidelity. Experiments show that ManipSynth achieves about $5\times$ higher interaction coverage and $6\times$ lower penetration than OMOMO across representative objects and interaction types. Using these synthetic demonstrations, we trained **ManipTracker**, a general-purpose HOI tracking controller with task-agnostic imitation rewards, reaching around 90% tracking success on training data while maintaining strong generalization on held-out interactions. Our current pipeline targets a limited interaction taxonomy and uses structured key states and rule-based transitions, which may under-cover highly unstructured scenarios (e.g., clutter, highly dynamic, multi-object interactions). Future work will expand interaction and object coverage, automate key-state/transition discovery, strengthen refinement under extreme dynamics, and better bridge to real data via domain randomization and sim-to-real calibration.

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Scaling Whole-Body Loco-Manipulation through Compositional Data Synthesis

Supplementary Material

A Simulation Setup

We use two closely related but not identical simulation configurations in our pipeline, since whole-body motion synthesis and downstream HOI tracking place different demands on physics fidelity and control frequency. For whole-body motion synthesis, the simulator mainly serves to generate scalable reference motions under stable locomotion and contact dynamics. For HOI tracking, the environment additionally needs to handle object interaction, contact-sensitive policy learning, and the training-time augmentation strategies described later in this supplement. Table 6 summarizes the core simulation hyperparameters used for whole-body motion synthesis, while Table 7 summarizes the simulation, contact-material, reward-weight, and initialization-disturbance settings used for HOI tracking training.

B HOI Tracking Training Details

Our HOI tracking training uses different stabilization strategies for large-object and small-object regimes. For large-object interactions, the main challenge is not only reference mismatch but also physical validity of the reset state: because the object is heavy and often requires sustained whole-body support or bi-manual coordination, small inconsistencies at initialization can quickly lead to penetration, unstable support contacts, or early rollout failure. We therefore adopt Physical State Initialization (PSI), which improves training by reusing physically feasible states collected from successful rollouts rather than relying exclusively on direct reference-state resets.

For small-object interactions, the failure mode is different. These motions are typically more sensitive to local hand articulation, object pose offsets, and transient contact perturbations than to long-horizon support instability. Accordingly, instead of emphasizing physically validated reset buffers, we apply online initialization disturbance during training. Concretely, we perturb a subset of initial states with randomized object pose, object velocity, and hand-state noise, so that the policy repeatedly learns to recover from nearby off-reference states and return to the target HOI trajectory. The detailed formulations of PSI and initialization disturbance are provided below after the simulation tables.

Table 6: Core simulator settings used for whole-body motion synthesis.

Setting	Choice
Physics integration step	1/60 s
Control update interval	1/30 s
Physics substeps per control interval	2
Physics solver	TGS
Position solver iterations	4
Velocity solver iterations	0
Contact offset	0.02
Rest offset	0.0
Bounce threshold velocity	0.2
Maximum depenetration velocity	10.0
Gravity direction and magnitude	(0, 0, -9.81)
Ground static / dynamic friction	1.0 / 1.0
Ground restitution	0.0

Physical State Initialization for large-object HOI tracking. We adopt Physical State Initialization (PSI) as a training strategy mainly for large-object HOI tracking. Compared with small-object manipulation, large-object interactions are substantially more sensitive to initialization quality: the object has larger inertia, often requires sustained arm-torso coordination or bi-manual support, and is more likely to exhibit contact mismatch if the episode starts from a kinematically plausible but physically inconsistent reference frame. As a result, directly applying reference-state initialization to synthesized trajectories can lead to immediate penetration, unstable support contacts, or rapid divergence between the humanoid and the object before the policy has a chance to recover.

To mitigate this issue, we maintain a buffer of physically validated rollout states collected during training. Each buffer entry stores a simulator state together with its aligned reference index,

$$\mathcal{B}_{\text{PSI}} = \{(\mathbf{s}_t, \kappa_t)\}, \quad (8)$$

where $\mathbf{s}_t$ contains the humanoid state, object state, and other simulator variables needed for reset, and κ_t records the associated demonstration sequence and frame index so that the future reference window remains temporally aligned after reset. During training, the initial state of an episode is sampled from a mixture of standard reference initialization and buffered physical states:

$$\mathbf{s}_0 \sim (1 - \rho) p_{\text{ref}} + \rho p_{\text{psi}}, \quad (9)$$

where p_{ref} denotes sampling from the demonstration frames and p_{psi} denotes sampling from $\mathcal{B}_{\text{PSI}}$. We apply PSI sparsely, namely only for a subset of resets, so that training still covers the original motion distribution while benefiting from physically feasible restart states.

The key update rule is that only high-quality rollout states are allowed to re-enter the initialization buffer. For a rollout fragment starting at frame t , we

compute its discounted return

$$G_t = \sum_{k=t}^T \gamma^{k-t} r_k. \quad (10)$$

States with sufficiently high G_t and without severe early failure are treated as valid physical states and inserted into $\mathcal{B}_{\text{PSI}}$, while old entries are discarded in a queue-like manner to keep the buffer compact and up to date. Intuitively, this mechanism bootstraps from successful partial interactions: once the policy discovers a physically coherent carrying or handover configuration for a large object, that state can later be reused as a training start point, making it much easier to practice subsequent phases such as transport, turning, and placement.

This strategy is particularly helpful for large objects because the most difficult portions of the motion often occur after contact has already been established. Without PSI, many episodes terminate early near the grasp-acquisition phase, which greatly reduces supervision for later-stage behaviors. PSI shifts part of the training distribution toward feasible post-contact states, allowing the policy to repeatedly rehearse long-horizon whole-body object transport under realistic contact and balance constraints. In our setting, PSI therefore acts as a practical bridge between imperfect synthetic references and stable large-object HOI policy optimization.

Initialization disturbance for small-object HOI tracking. For small-object HOI tracking, we additionally use online data augmentation by perturbing the initial state at reset time. The goal is not to create arbitrary corruptions, but to expose the policy to nearby off-reference states from which it must recover and rejoin the original demonstration. This is especially useful for small-object skills because the object pose, hand articulation, and contact condition can change rapidly over short time horizons, making the controller sensitive to small initialization errors.

We use a Bernoulli-gated noise operator

$$\epsilon(\sigma, p) = z \odot u, \quad z \sim \text{Bernoulli}(p), \quad u \sim \mathcal{U}([- \sigma, \sigma]), \quad (11)$$

where the binary variable z is sampled per environment instance and determines whether that reset is perturbed. Thus, only a subset of training episodes is disturbed, while the others remain on the original reference distribution. At the object level, we perturb the initial linear and angular velocities as

$$\tilde{\mathbf{v}}_0^o = \mathbf{v}_0^o + \epsilon(0.02, p), \quad \tilde{\boldsymbol{\omega}}_0^o = \boldsymbol{\omega}_0^o + \epsilon(0.02, p). \quad (12)$$

We then apply contact-conditioned pose perturbations. When the initial frame is non-contact, we use a larger translational disturbance and allow full 3-axis rotational perturbation:

$$\tilde{\mathbf{p}}_0^o = \mathbf{p}_0^o + \epsilon(0.05, p), \quad \tilde{\mathbf{q}}_0^o = \mathcal{R}_{xyz}(\epsilon(\pi/5, p)) \otimes \mathbf{q}_0^o, \quad (13)$$

with the additional constraint that the vertical position disturbance is upward-only. When the object is already in contact, we use a smaller positional perturbation and restrict orientation noise to yaw:

$$\tilde{\mathbf{p}}_0^o = \mathbf{p}_0^o + \epsilon(0.02, p), \quad \tilde{\mathbf{q}}_0^o = \mathcal{R}_z(\epsilon(\pi/8, p)) \otimes \mathbf{q}_0^o. \quad (14)$$

This asymmetric design reflects the fact that disturbing a contacted small object too aggressively can immediately destroy the grasp geometry, whereas non-contact frames permit broader perturbation.

On the human side, we perturb the hand articulation states while leaving the rest of the body initialization unchanged. In practice, we add random offsets to the left- and right-hand DoF positions and velocities:

$$\tilde{\mathbf{q}}_0^{h,\text{hand}} = \mathbf{q}_0^{h,\text{hand}} + \epsilon(\pi/8, p), \quad \dot{\tilde{\mathbf{q}}}_0^{h,\text{hand}} = \dot{\mathbf{q}}_0^{h,\text{hand}} + \epsilon(0.1, p). \quad (15)$$

These disturbances encourage the policy to correct small finger configuration errors and object-state deviations at the beginning of an episode. Overall, this reset-time augmentation substantially enriches the effective training distribution for small-object manipulation and improves robustness to imperfect grasps, pose offsets, and transient contact inconsistencies.

C HOI tracking reward detail

Following Eq. (7) in the main paper, we use a unified imitation reward for human-object interaction (HOI) tracking:

$$r_t = r_t^h \cdot r_t^o \cdot r_t^i \cdot r_t^c \cdot r_t^f. \quad (16)$$

The reward is evaluated after each simulator transition using the current simulated state and the aligned reference frame. For brevity, we omit the one-step index shift and write all quantities at time t . Let $\mathcal{B}$ denote the set of tracked humanoid rigid bodies, $\mathcal{K}$ the set of canonical object keypoints, and $\mathcal{C}$ the set of human links that are allowed to interact with the object. We use $(\mathbf{p}_t^b, \mathbf{R}_t^b)$ and $(\hat{\mathbf{p}}_t^b, \hat{\mathbf{R}}_t^b)$ for the simulated and reference pose of body $b \in \mathcal{B}$, and $(\mathbf{p}_t^o, \mathbf{R}_t^o)$ and $(\hat{\mathbf{p}}_t^o, \hat{\mathbf{R}}_t^o)$ for the simulated and reference object pose. All imitation sub-rewards share the same exponential form

$$r_t^\alpha = \exp(-\lambda_\alpha e_t^\alpha), \quad (17)$$

where e_t^α is the corresponding tracking error and $\lambda_\alpha > 0$ is a fixed weight.

Human tracking reward. The human term encourages the humanoid to match the reference whole-body motion in both global body positions and orientations:

$$r_t^h = r_t^{hp} \cdot r_t^{hr}, \quad (18)$$

$$e_t^{hp} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \|\mathbf{p}_t^b - \hat{\mathbf{p}}_t^b\|_2^2, \quad (19)$$

$$e_t^{hr} = \frac{1}{|\mathcal{B}|} \sum_{b \in \mathcal{B}} \left\| \log \left(\mathbf{R}_t^b \hat{\mathbf{R}}_t^{b\top} \right) \right\|_2^2. \quad (20)$$

Here $\log(\cdot)$ denotes the rotation logarithm map, so e_t^{hr} measures geodesic orientation mismatch on $\text{SO}(3)$. Compared with hand-centric trackers, this term supervises the entire humanoid kinematic chain and is therefore crucial for preserving locomotion rhythm, balance, torso coordination, and arm posture simultaneously.

Object tracking reward. The object term aligns the rigid object motion with the demonstration:

$$r_t^o = r_t^{op} \cdot r_t^{or}, \quad (21)$$

$$e_t^{op} = \left\| \mathbf{p}_t^o - \hat{\mathbf{p}}_t^o \right\|_2^2, \quad (22)$$

$$e_t^{or} = \left\| \log \left(\mathbf{R}_t^o \hat{\mathbf{R}}_t^{o\top} \right) \right\|_2^2. \quad (23)$$

This term ensures that the tracked policy follows the demonstrated object transport, placement, and reorientation behavior instead of only matching the human motion.

Interaction reward. Accurate HOI tracking additionally requires preserving the relative geometry between the human and the object. To this end, we define a set of interaction pairs $\mathcal{P} \subseteq \mathcal{B} \times \mathcal{K}$ between interaction-relevant human body parts and object keypoints. For an object keypoint $\bar{\mathbf{k}}^m$ in the object local frame, its world-space position is

$$\mathbf{k}_t^m = \mathbf{R}_t^o \bar{\mathbf{k}}^m + \mathbf{p}_t^o. \quad (24)$$

We then measure the consistency of human-object relative vectors:

$$\mathbf{d}_t^{b,m} = \mathbf{k}_t^m - \mathbf{p}_t^b, \quad \hat{\mathbf{d}}_t^{b,m} = \hat{\mathbf{k}}_t^m - \hat{\mathbf{p}}_t^b, \quad (25)$$

$$e_t^i = \frac{1}{|\mathcal{P}|} \sum_{(b,m) \in \mathcal{P}} \left\| \mathbf{d}_t^{b,m} - \hat{\mathbf{d}}_t^{b,m} \right\|_2^2, \quad r_t^i = \exp(-\lambda_i e_t^i). \quad (26)$$

This is the whole-body HOI analogue of relative hand-object tracking used in dexterous manipulation. It stabilizes the demonstrated coupling between the object and the relevant human links, especially during grasp acquisition, carrying, handover, and release, where small absolute errors can accumulate into large interaction failures.

Contact reward. In our final implementation, the contact term only supervises whether the object is in contact, rather than matching per-link human contact patterns. Let $c_t^o \in \{0, 1\}$ denote the simulated object-contact indicator extracted from the current observation, and let $\hat{c}_t^o$ denote the reference object-contact labels. Following the code, we first collapse the reference contact vector into a single binary target

$$\hat{c}_t^o = \mathbb{I}[\max \hat{c}_t^o > 0.1]. \quad (27)$$

The resulting reward compares the simulated object-contact state against this binary reference target:

$$e_t^c = |c_t^o - \hat{c}_t^o|, \quad r_t^c = \exp(-\lambda_c e_t^c), \quad (28)$$

where λ_c corresponds to the configured object-contact weight (`cg_obj`). This term complements the interaction-geometry reward by explicitly distinguishing contact and non-contact phases at the object level, which is sufficient in our setting to stabilize grasp acquisition, object transport, and release without imposing additional human-link contact supervision.

Feet-in-the-air-time reward. We keep the implementation of the Isaac Gym `feet_air_time` reward unchanged. Let $\mathcal{F}$ denote the set of foot bodies used for locomotion supervision. For each foot $j \in \mathcal{F}$, we first define a binary contact indicator from the vertical contact force:

$$c_t^j = \mathbb{I}\left[f_{t,z}^j > 1\right], \quad (29)$$

where $f_{t,z}^j$ is the z -component of the contact force on foot j . Since mesh-based contact reports can be noisy, the implementation applies a one-step temporal filter

$$\tilde{c}_t^j = c_t^j \vee c_{t-1}^j, \quad (30)$$

where $\vee$ denotes the logical OR. Let a_t^j be the accumulated air time of foot j before the current update. A foot is considered to make a valid touchdown only if it was previously airborne:

$$b_t^j = \mathbb{I}\left[a_t^j > 0\right] \tilde{c}_t^j. \quad (31)$$

The code then increments the air-time buffer by the simulation step Δt and evaluates reward only on this first filtered contact:

$$r_t^f = \sum_{j \in \mathcal{F}} \left(a_t^j + \Delta t - 0.25\right) b_t^j. \quad (32)$$

Finally, the air-time state is reset for contacting feet and continued for airborne feet:

$$a_{t+1}^j = \left(a_t^j + \Delta t\right) \left(1 - \tilde{c}_t^j\right). \quad (33)$$

This reward is therefore issued only at touchdown events after a non-zero swing phase, and it favors longer, cleaner steps while suppressing frequent short contacts and foot-skating artifacts during locomotion-heavy HOI tracking.

D Limitations and Future Work

Although, for the same object instances, our synthesized data achieve substantially higher interaction coverage and lower penetration than OMOMO, the current framework still has several important limitations. First, the interaction patterns that can be generated reliably remain relatively narrow. Our current compositional pipeline is effective for structured modes such as pick-and-place and handover, but it is less straightforward to synthesize whole-body motion for interactions whose body-object coupling is dominated by sustained external force and scene-dependent contact transitions, such as pushing or dragging. A promising future direction is to incorporate single-frame human-object interaction snapshots from mocap data as anchors, and then perform data augmentation under approximately preserved human-object relative configurations, so as to seed more diverse interaction families without requiring full motion capture sequences. Second, the present model is tailored to rigid objects. Consequently, it does not yet support manipulation involving deformable materials or fluids, such as folding garments, where the object state is no longer well captured by a single rigid-body pose and contact evolution becomes substantially more complex. Addressing this limitation will likely depend on advances in simulation backends that can represent deformable and fluid interactions with sufficient fidelity and efficiency for large-scale policy learning. Third, our sparse whole-body motion generation stage inherits the foot-sliding behavior that can arise in MaskedMimic-style tracking. One plausible reason is that MaskedMimic relies primarily on a two-stage supervised pipeline and lacks a final stage with real environment interaction, making it difficult to correct contact-timing and locomotion errors that only become apparent under physics. A natural next step is therefore to build on the existing second-stage model with an additional reinforcement learning fine-tuning stage, which may help reduce residual foot skating and improve physical consistency in the generated whole-body motions.

E Additional Demo Videos

Beyond the representative qualitative results included in the paper and this supplementary document, we provide a larger collection of demo videos in the accompanying HTML file. These videos are intended to give a broader view of the method’s behavior across different tasks, object scales, and evaluation settings. In particular, the HTML file contains additional examples of sparse whole-body motion synthesis from different tracking-input configurations, more ManipSynth results spanning both small-object and large-object interactions such as handover and pick-and-place, and side-by-side ManipSynth/ManipTracker comparisons for representative large-object HOI sequences.

Table 7: Simulation, contact-material, reward-weight, and initialization-disturbance settings used for HOI tracking training.

Setting	Choice
Physics integration step	1/120 s
Policy action interval	1/30 s
Physics solver	TGS
Position solver iterations	4
Velocity solver iterations	1
Continuous collision detection	Enabled
Humanoid contact offset	0.02
Humanoid rest offset	0.0
Humanoid maximum depenetration velocity	100.0
Object density	200
Object static / dynamic friction	0.6 / 0.6
Object restitution	0.05
Ground static / dynamic friction	0.9 / 0.9
Ground restitution	0.1
Object collision approximation	Convex decomposition
Maximum convex hulls per object	64
Reward Weights	
Human position reward weight (p)	30.0
Human rotation reward weight (r)	2.5
Human position-velocity reward weight (pv)	0.0
Human angular-velocity reward weight (rv)	0.0
Object position reward weight (op)	5.0
Object rotation reward weight (or)	0.1
Object position-velocity reward weight (opv)	0.1
Object angular-velocity reward weight (orv)	0.0
Interaction-geometry reward weight (ig)	5.0
Object-contact reward weight (cg_obj)	5.0
Human motion energy weight (eg1)	2.0×10^{-5}
Initialization Disturbance (Small-Object HOI)	
Initialization disturbance probability (state_noise_prob)	0.2
Object linear / angular velocity noise scale	0.02 / 0.02
Non-contact object position noise scale	0.05
Non-contact object rotation range	$\pi/5$
Contact object position noise scale	0.02
Contact object yaw-noise range	$\pi/8$
Hand DoF position noise scale	$\pi/8$
Hand DoF velocity noise scale	0.1